The iterates of a self-referential power series modulo 4

A proof of four conjectures in OEIS A396099

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Abstract

Let $A(x) = x + x^2 + \cdots$ be the power series defined by the equation $A(x) = x + A^{\circ 2}(x) \cdot A^{\circ 3}(x)$, where $A^{\circ k}$ means A substituted into itself k times. Its coefficients form the sequence A396099 of the On-Line Encyclopedia of Integer Sequences (OEIS). Paul D. Hanna conjectured that every $a(n)$ is odd, that from $n = 3$ on the remainders of $a(n)$ on division by 4 repeat the block $1, 3, 3, 1$, that from $n = 3$ on the coefficients of $A(A(x))$ are divisible by 4, and that from $n = 3$ on the coefficients of $A(x) - x\,A(A(A(x)))$ leave remainder 2 on division by 4. We prove all four. The key fact is that, modulo 4, the series A equals the explicit fraction $\frac{x}{1-x} + \frac{2x^4}{(1-x)^3}$, and $A(A(x))$ equals $x + 2x^2$. The proof covers the whole family of equations $F = x + F^{\circ r} \cdot F^{\circ(r+1)}$, $r \geq 1$. Modulo $2r$, every iterate $F^{\circ k}$ (including those with $k < 0$, built from the inverse series) equals the fraction $x/(1 - kx)$, plus an explicit correction $r\,x^4/(1-x)^3$ when r is even and k is odd. All results have been checked by computer and proved formally in the Lean proof assistant.

How this paper was produced. This paper reports the result of an AI research effort directed by Dev D. Goyal. ChatGPT (OpenAI), using the GPT-6 model with Pro reasoning and working autonomously from prompts by Dev D. Goyal, found the proof, wrote its Lean 4 formalization, and wrote a first draft of the paper. Dev D. Goyal chose the problems and asked for proofs, formal verification and revisions, but did not supply proof ideas. Claude Opus 5.5, used with Claude Code, then checked the work independently: it rebuilt the formal proofs from source and confirmed that they rely only on Lean's three standard axioms, recomputed the sequence from its definition compared it with the published terms and tested every claim numerically, and read every step of the proofs. It also wrote the present version of this paper. Dev D. Goyal has verified the results and is responsible for them. The paper has not been peer reviewed. The section "About this paper" at the end gives the details.

1 Introduction

1.1 The sequence

The *On-Line Encyclopedia of Integer Sequences* (OEIS) [4] is a large public database of integer sequences. Many entries also record *conjectures*: patterns that the contributor saw in the data but did not prove.

This paper is about entry A396099 [1], contributed by Paul D. Hanna in June 2026. The entry defines a power series $A(x) = \sum_{n \geq 1} a(n)\,x^n$ by an equation in which A is substituted into itself. The entry reads:

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G.f.  satisfies A(x) = x + A^2(x)*A^3(x).
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Conjecture:  all terms are odd.

Conjecture:  a(n) = [1,3,3,1] repeating (mod 4) for n > 2.

Conjecture:  [x^n] A(A(x)) == 0 (mod 4) for n > 2.

Conjecture:  [x^n] ( A(x) - x*A(A(A(x))) ) == 2 (mod 4) for n > 2.
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— OEIS A396099, Paul D. Hanna, June 2026

Here "G.f." stands for *generating function*, the series whose coefficients are the terms of the sequence. The notation needs care. In the entry, $A^k(x)$ does not mean the k-th power of $A(x)$, but the k-th *iterate*: A substituted into itself k times, so that $A^2(x) = A(A(x))$ and $A^3(x) = A(A(A(x)))$. The entry says so explicitly in its list of formulas ("$A^n(x)$ denotes the n-th iteration of $A(x)$"). To avoid any confusion we write $A^{\circ k}$ for the iterate and keep A^k for the ordinary power. So the sequence is defined by

$$A(x) = x + A^{\circ 2}(x) \cdot A^{\circ 3}(x), \qquad A(x) = x + a(2)x^2 + a(3)x^3 + \cdots, \tag{1}$$

and $a(n)$ is the coefficient of x^n in $A(x)$.

At first sight it is not clear that (1) defines anything, since the unknown series appears on both sides, substituted into itself many times. It does: every iterate of A starts with x, so the product on the right starts with x^2; this gives $a(2) = 1$. In general the coefficient of x^n on the right only involves $a(1), \ldots, a(n-1)$, so each new coefficient is determined by the earlier ones (Lemma 4.1 below makes this precise). The first terms, starting at $n = 1$, are 1, 1, 5, 39, 383, 4357, 55057, 754223, 11028735, ...; they grow quickly.

1.2 The question

The entry makes four conjectures. Here $[x^n]\,U$ denotes the coefficient of x^n in a series U, and "==" or "$= \ldots \pmod 4$" means that the two sides leave the same remainder on division by 4. In other words,

(a) every $a(n)$ is odd;

(b) from $n = 3$ on, the remainders of $a(n)$ on division by 4 repeat the block $1, 3, 3, 1$: the remainders of $a(3), a(4), a(5), a(6)$ are $1, 3, 3, 1$, those of $a(7), \ldots, a(10)$ are again $1, 3, 3, 1$, and so on;

(c) from x^3 on, every coefficient of the second iterate $A(A(x))$ is divisible by 4;

(d) from x^3 on, every coefficient of $A(x) - x \cdot A(A(A(x)))$ leaves remainder 2 on division by 4.

The table shows the data. The indices $n \geq 3$, where the patterns are claimed, are printed in bold.

n	1	2	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**	**11**	**12**	**13**	**14**	**15**	**16**
$a(n) \bmod 4$	1	1	**1**	**3**	**3**	**1**	**1**	**3**	**3**	**1**	**1**	**3**	**3**	**1**	**1**	**3**
$[x^n]\,A^{\circ 2} \bmod 4$	1	2	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
$[x^n]\,(A - xA^{\circ 3}) \bmod 4$	1	0	**2**	**2**	**2**	**2**	**2**	**2**	**2**	**2**	**2**	**2**	**2**	**2**	**2**	**2**

n	**17**	**18**	**19**	**20**	**21**	**22**	**23**	**24**	**25**	**26**	**27**	**28**	**29**	**30**	**31**	**32**
$a(n) \bmod 4$	**3**	**1**	**1**	**3**	**3**	**1**	**1**	**3**	**3**	**1**	**1**	**3**	**3**	**1**	**1**	**3**
$[x^n]\,A^{\circ 2} \bmod 4$	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
$[x^n]\,(A - xA^{\circ 3}) \bmod 4$	**2**	**2**	**2**	**2**	**2**	**2**	**2**	**2**	**2**	**2**	**2**	**2**	**2**	**2**	**2**	**2**

1.3 What this paper proves

All four conjectures are true (Corollary 3.2). The reason is that, modulo 4, the series A equals the explicit series

$$\frac{x}{1-x} + \frac{2x^4}{(1-x)^3} = x + x^2 + x^3 + 3x^4 + 7x^5 + 13x^6 + \cdots,$$

and its second iterate $A(A(x))$ is $x + 2x^2$. The four patterns can be read off from these formulas.

More generally, we treat the whole family of equations

$$F = x + F^{\circ r} \cdot F^{\circ(r+1)}, \qquad r = 1, 2, 3, \ldots,$$

of which the entry is the case $r = 2$. Theorem 3.1 gives, for every r, every iterate $F^{\circ k}$ of the solution modulo $2r$, for every integer k, positive, zero or negative. Up to an explicit correction when r is even, F is congruent modulo $2r$ to the fraction $x/(1-x)$, whose k-th iterate is $x/(1-kx) = x + kx^2 + k^2x^3 + \cdots$. In particular, modulo $2r$, substituting F into itself $2r$ times gives back x, and no smaller number of substitutions does.

1.4 What was known

Hanna recorded the conjectures in June 2026, as part of a family of four entries with $r = 2, 3, 4, 5$: A396099, A396797, A396798 and A396807. For $r = 3$ (entry A396797) the conjectures were proved by Shiva Priyan Shanmuga Sundaram in September 2026 [2]; the proof is linked from that entry. Our argument for odd r compares the solution with the series $x/(1-x)$ and uses the uniqueness of the solution, for every odd r at once. For even r, as here, the comparison with $x/(1-x)$ alone does not work; the correction term rH in Theorem 3.1 is the extra ingredient that even r needs. When we checked the entries on 24 October 2026, the statements for $r = 2, 4, 5$ were still labelled as conjectures. We are not aware of an earlier proof of them.

1.5 Motivation

Most equations that define sequences in the OEIS combine *known* series by sums, products and substitutions. In (1) the unknown series is substituted into *itself*, several times, inside its own definition. No closed formula for A is known, and its coefficients grow quickly; Hanna's conjectures record simple patterns in their remainders.

The explanation given here is that, modulo $2r$, the series A is congruent to the fraction $x/(1-x)$, up to an explicit correction term. Iterates of power series appear throughout combinatorics and in the study of dynamical systems; here every iterate, including those with negative index, is determined modulo $2r$. The method is general: find a candidate solution modulo a small integer, verify that it satisfies the reduced equation, and conclude by uniqueness. The correction needed for even r uses a second general idea, a Taylor formula that is exact because the correction squares to zero.

1.6 How to read this paper

Section 2 explains the few notions we need: power series, substitution and iteration, inverse series, and remainders. Section 3 states the results precisely, and Section 4 proves them; the proof of the main theorem comes first, and the consequences for this entry are derived from it

at the end of that section. Section 5 describes how the results were checked, by computer and by a formal proof in Lean. Section 6 discusses the scope of the results, open questions, and related work.

2 Background

This section collects the definitions and facts used later. Readers who know formal power series and congruences can skip it and return when needed.

2.1 Formal power series

A *formal power series* is an expression

$$U(x) = u_0 + u_1 x + u_2 x^2 + u_3 x^3 + \cdots,$$

determined by its sequence of coefficients $u_0, u_1, u_2, \ldots$; see [6, 5]. No number is ever substituted for x, so questions of convergence do not arise. We write $[x^n]\,U = u_n$ for the coefficient of x^n in U; two series are equal when all their coefficients are equal.

Series are added coefficient by coefficient and multiplied like polynomials:

$$[x^n]\,(UV) = u_0 v_n + u_1 v_{n-1} + \cdots + u_n v_0.$$

Each coefficient of a product is a finite sum. For example, $(1-x)(1+x+x^2+x^3+\cdots) = 1$, so $1+x+x^2+\cdots$ is the *multiplicative inverse* of $1-x$, written $\frac{1}{1-x}$. More generally, every series D with constant term 1 has a unique multiplicative inverse $\frac{1}{D}$, whose coefficients are determined one at a time by $D \cdot \frac{1}{D} = 1$ using only addition, subtraction and multiplication. Whenever a quotient of power series appears in this paper, its denominator has constant term 1, so the quotient is again a power series.

A *commutative ring* is a set with an addition, a subtraction and a multiplication that satisfy the usual rules of arithmetic; examples are $\mathbb{Z}$, $\mathbb{Q}$ and the integers modulo m. For a commutative ring R we write $R[[x]]$ for the set of power series with coefficients in R; for instance, writing $\mathbb{F}_2$ for the integers modulo 2, $\mathbb{F}_2[[x]]$ consists of the series with coefficients modulo 2. We write $U(0)$ for the constant term u_0 of U, and $U = O(x^d)$ when the coefficients of $1, x, \ldots, x^{d-1}$ in U are all zero. Thus $U - V = O(x^d)$ means that U and V have the same coefficient of x^n for every $n < d$. A series with constant term 0 and coefficient of x equal to 1, that is, a series of the form $x + u_2 x^2 + u_3 x^3 + \cdots$, is called *normalized*.

2.2 Substituting one series into another

If $V(x)$ has constant term 0, we can substitute V for x in U:

$$U(V(x)) = u_0 + u_1 V(x) + u_2 V(x)^2 + u_3 V(x)^3 + \cdots.$$

This is called *composition*, also written $U \circ V$. It makes sense because $V(x)^k$ has no terms below x^k, so only the finitely many terms with $k \le n$ affect the coefficient of x^n. For example, if $U = x + x^2$ and $V = x^2$, then $U(V(x)) = x^2 + x^4$. Composition is associative: $U \circ (V \circ W) = (U \circ V) \circ W$.

Substituting a series into itself repeatedly gives its *iterates*: $U^{\circ 2}(x) = U(U(x))$, $U^{\circ 3}(x) = U(U(U(x)))$, and so on, with $U^{\circ 0}(x) = x$ and $U^{\circ 1} = U$. They must not be confused with

powers $U^2 = U \cdot U$, $U^3 = U \cdot U \cdot U$. (The OEIS sometimes writes $A^k(x)$ for the k-th iterate. In this paper the small circle always means an iterate, and a plain exponent always means a power.)

2.3 Undoing a substitution

A normalized series can always be "undone" by another normalized series.

Lemma 2.1 (Compositional inverse)**.** *For every normalized series U there is exactly one normalized series V with $U(V(x)) = x$, and this V also satisfies $V(U(x)) = x$. If U has integer coefficients, so does V.*

We call V the *compositional inverse* of U. The OEIS calls it the *series reversion* of U. For example, the compositional inverse of $x - x^2$ is $C(x) = x + x^2 + 2x^3 + 5x^4 + 14x^5 + \cdots$, whose coefficients are the Catalan numbers. Indeed $C - C^2 = x$, which says exactly that substituting C into $x - x^2$ gives x.

Proof. Write $V = x + v_2x^2 + v_3x^3 + \cdots$ with unknown coefficients. In $U(V) = V + u_2V^2 + u_3V^3 + \cdots$, the coefficient of x^n (for $n \geq 2$) equals v_n plus terms that involve only $v_2, \ldots, v_{n-1}$. Setting it equal to 0 therefore determines v_n from the earlier coefficients, using only additions, subtractions and multiplications. So there is exactly one V with $U \circ V = x$, and it has integer coefficients if U does. Applying the same construction to V gives a normalized W with $V \circ W = x$. Then $U = U \circ (V \circ W) = (U \circ V) \circ W = W$, so $V \circ U = V \circ W = x$ as well. □

2.4 Iterates with negative index, and the order of a series

Let F be a normalized series and let F^{-1} be its compositional inverse (Lemma 2.1). For a negative integer k we define $F^{\circ k}$ as the $|k|$-th iterate of F^{-1}; for example $F^{\circ(-2)} = F^{-1}(F^{-1}(x))$. With this convention $F^{\circ j} \circ F^{\circ k} = F^{\circ(j+k)}$ for all integers j and k, just as for powers of a number. The *order* of F under composition is the smallest $k \geq 1$ with $F^{\circ k}(x) = x$, if there is one. Over the integers only $F = x$ has finite order, but after reduction modulo m other series can have finite order, and this will happen below.

We also use two pieces of notation. For a statement S we write $[S]$ for 1 if S is true and 0 if it is false (for example $[k$ odd$]$). And $\binom{j}{2} = j(j-1)/2$ is the number of ways to choose 2 objects out of j.

2.5 Remainders and congruences

For an integer $m \geq 2$ we write $a \equiv b \pmod{m}$ ("a is congruent to b modulo m") when $a - b$ is divisible by m, that is, when a and b leave the same remainder after division by m. The remainder of a on division by m is called its *residue* modulo m; it is one of $0, 1, \ldots, m-1$.

Addition and multiplication are compatible with congruences, so the residues $0, 1, \ldots, m-1$, added and multiplied modulo m, form a ring, written $\mathbb{Z}/m\mathbb{Z}$. When $m = p$ is a prime we also write $\mathbb{F}_p$. In $\mathbb{F}_2 = \{0, 1\}$ we have $1 + 1 = 0$, so addition and subtraction coincide.

For power series with integer coefficients, $U \equiv V \pmod{m}$ means that *every* coefficient of $U - V$ is divisible by m. Equivalently, the two series become equal when every coefficient is replaced by its residue; we call this *reducing* the series modulo m. The key fact is the following.

Reduction modulo m is compatible with all the operations above. To reduce a sum, a product, a substitution, or an inverse, we may reduce the ingredients first and then compute in $\mathbb{Z}/m\mathbb{Z}$.

This holds because every coefficient of the result is obtained from finitely many coefficients of the ingredients by adding, subtracting and multiplying, and these operations are compatible with remainders.

Rings. Some statements below hold over every commutative ring, for example $\mathbb{Z}$, $\mathbb{Q}$ and $\mathbb{Z}/m\mathbb{Z}$. No division is used in them; this matters because in $\mathbb{Z}/m\mathbb{Z}$ division is not always possible (for instance, 2 has no inverse in $\mathbb{Z}/4\mathbb{Z}$).

2.6 The method

The main proofs follow three steps.

1. *Uniqueness.* Show that the defining equation determines the coefficients one at a time, using only addition, subtraction and multiplication. Then the equation has exactly one solution of the required shape, in every commutative ring, including $\mathbb{Z}/2r\mathbb{Z}$.
2. *Reduction.* Reducing the integer solution modulo $2r$ gives a solution of the reduced equation, because reduction is compatible with all operations.
3. *A candidate solution.* Find an explicit series that satisfies the reduced equation. By uniqueness, it is the reduction of the integer solution, and its coefficients can be read off.

Finding the candidate in step 3 is usually the main difficulty.

2.7 A one-term Taylor formula

Lemma 2.2. *Let ε be an element of a commutative ring with $\varepsilon^2 = 0$. If u and v are series with constant term 0, then for every series S*

$$S(u + \varepsilon v) = S(u) + \varepsilon\, v\, S'(u),$$

where $S' = \sum_{n\geq 1} n\, s_n x^{n-1}$ is the formal derivative of S.

Proof. By the binomial theorem, $(u+\varepsilon v)^j = u^j + j\,\varepsilon v\, u^{j-1}$, because all terms with ε^2 or higher powers vanish. Multiply by s_j and add up over j. □

An example of such an ε: in $\mathbb{Z}/8\mathbb{Z}$, the element 4 satisfies $4^2 = 16 = 0$.

3 Results

For every integer $r \geq 1$, Lemma 4.1 below shows that there is exactly one normalized series F_r with

$$F_r = x + F_r^{\circ r} \cdot F_r^{\circ (r+1)}, \tag{2}$$

and that it has integer coefficients. The series of the entry is $A = F_2$. To state the result we use two simple series with integer coefficients. For an integer k, let

$$T_k(x) = \frac{x}{1-kx} = x + kx^2 + k^2x^3 + \cdots = \sum_{n\geq 1} k^{\,n-1} x^n,$$

with the convention $k^0 = 1$ (so $T_0 = x$), and let

$$H(x) = \frac{x^4}{(1-x)^3} = x^4 + 3x^5 + 6x^6 + 10x^7 + \cdots = \sum_{n\geq 4} h_n\, x^n, \qquad h_n = \binom{n-2}{2}.$$

We also set $h_n = 0$ for $n < 4$. (Lemma 4.2 checks the formula for h_n.)

Theorem 3.1. *Let $r \geq 1$ and let k be any integer. Modulo $2r$,*

$$F_r^{\circ k} \equiv \begin{cases} T_k + r\,H, & \text{if } r \text{ is even and } k \text{ is odd,} \\ T_k, & \text{otherwise.} \end{cases}$$

In terms of coefficients: for every $n \geq 1$,

$$[x^n]\,F_r^{\circ k} \equiv k^{\,n-1} + r\,h_n\,[\,r \text{ even and } k \text{ odd}] \pmod{2r}.$$

Moreover, $F_r^{\circ k} \equiv x \pmod{2r}$ if and only if k is a multiple of $2r$. So, modulo $2r$, the series F_r has order exactly $2r$ under composition.

In other words, when r is odd, the k-th iterate of F_r has, modulo $2r$, the same coefficients as $x/(1-kx)$, namely the powers $1, k, k^2, k^3, \ldots$. When r is even, the same is true for even k, while for odd k the coefficients are shifted by $r\,h_n$; since $2r \equiv 0 \pmod{2r}$, the shift $r\,h_n$ depends only on the parity of h_n.

For the entry, the theorem gives the following.

Corollary 3.2. *For $A = F_2$, the series of OEIS A396099, modulo 4:*

(a) $a(n)$ is odd for every $n \geq 1$;

(b) for $n \geq 3$, $a(n) \equiv 1$ if $n \equiv 2$ or $3 \pmod 4$, and $a(n) \equiv 3$ if $n \equiv 0$ or $1 \pmod 4$; so the remainders run $1, 3, 3, 1, 1, 3, 3, 1, \ldots$ from $n = 3$ on;

(c) $A^{\circ 2} \equiv x + 2x^2 \pmod 4$, so $[x^n]\,A(A(x)) \equiv 0 \pmod 4$ for every $n \geq 3$;

(d) $A - x\,A^{\circ 3} \equiv x + \dfrac{2x^3}{1-x} \pmod 4$, so $[x^n]\,(A(x) - x\,A(A(A(x)))) \equiv 2 \pmod 4$ for every $n \geq 3$.

These are exactly the four conjectures of the entry. Because the equation has only one normalized solution (Lemma 4.1), the four statements hold for every normalized series with integer coefficients that satisfies (1); the Lean formalization states them in that form as well.

4 Proofs

The idea in brief. We follow the method of Section 2. First we show that (2) has exactly one solution, over every commutative ring, so it is enough to find *some* solution modulo $2r$. For odd r, the fraction $T_1 = x/(1-x)$ works: its iterates are $T_k = x/(1-kx)$, and a one-line calculation shows that T_1 satisfies the equation modulo $2r$. For even r, T_1 does not satisfy the equation, and we correct it to $T_1 + rH$. Because r^2 is a multiple of $2r$, every calculation with the correction can be done "to first order" with the Taylor formula of Lemma 2.2, which is exact here. The statements of the entry are then read off from the formula.

4.1 Existence and uniqueness

Lemma 4.1. *Let R be a commutative ring and $r \geq 1$. There is exactly one normalized series F with coefficients in R such that $F = x + F^{\circ r} \cdot F^{\circ (r+1)}$. Its coefficients are obtained from those of lower degree using only additions and multiplications. In particular (2) defines a series F_r with integer coefficients, and for every $m \geq 2$ the reduction of F_r modulo m is the unique solution over $\mathbb{Z}/m\mathbb{Z}$.*

Proof. Write $\Phi(F) = x + F^{\circ r} \cdot F^{\circ(r+1)}$ for the right-hand side. The key observation is:

If two normalized series U and V agree below degree N (that is, $U - V = O(x^N)$, with $N \geq 2$), then $\Phi(U)$ and $\Phi(V)$ agree below degree $N + 1$.

To see this, we first show that $U^{\circ k} - V^{\circ k} = O(x^N)$ for every $k \geq 0$, by induction on k. The case $k = 0$ is clear. For the step, write

$$U^{\circ(k+1)} - V^{\circ(k+1)} = \big(U(U^{\circ k}) - U(V^{\circ k})\big) + \big(U(V^{\circ k}) - V(V^{\circ k})\big).$$

The first bracket is a sum of terms $u_j\big((U^{\circ k})^j - (V^{\circ k})^j\big)$, and each of them is divisible by $U^{\circ k} - V^{\circ k} = O(x^N)$. The second bracket is the series $U - V = O(x^N)$ with $V^{\circ k} = x + \cdots$ substituted for x, so it is $O(x^N)$ as well. Next, since all iterates start with x,

$$U^{\circ r}U^{\circ(r+1)} - V^{\circ r}V^{\circ(r+1)} = \big(U^{\circ r} - V^{\circ r}\big)\,U^{\circ(r+1)} + V^{\circ r}\,\big(U^{\circ(r+1)} - V^{\circ(r+1)}\big),$$

and each product is $O(x^N)$ times a series starting with x, hence $O(x^{N+1})$. This proves the observation.

Existence. Start with $G_0 = x$ and put $G_{i+1} = \Phi(G_i)$. Each G_i is normalized. The series $G_0 = x$ and $G_1 = x + x^2 + \cdots$ agree below degree 2, and by the observation and induction, G_i and G_{i+1} agree below degree $i + 2$. So the coefficient of x^n in G_i no longer changes once $i \geq n - 1$, and we let F be the series whose coefficient of x^n is this final value. Then F and G_i agree below degree $i + 2$, so $\Phi(F)$ and $\Phi(G_i) = G_{i+1}$ agree below degree $i + 3$; hence $\Phi(F)$ and F agree below degree $i + 2$ for every i, that is, $\Phi(F) = F$.

Uniqueness. If U and V are two normalized solutions, they agree below degree 2. If they agree below degree N, then $U = \Phi(U)$ and $V = \Phi(V)$ agree below degree $N + 1$. So they agree in every degree.

The construction uses only additions and multiplications of coefficients, so it works over every commutative ring and gives integer coefficients over $\mathbb{Z}$. Finally, reducing (2) modulo m shows that the reduction of F_r is a normalized solution over $\mathbb{Z}/m\mathbb{Z}$, because reduction is compatible with products and substitutions; by uniqueness it is the only one. □

4.2 Two calculations with fractions

Lemma 4.2. *Over every commutative ring, $T_c \circ T_d = T_{c+d}$ for all c, d. Consequently $T_1^{\circ k} = T_k$ for every integer k, $T_0 = x$, and T_{-k} is the compositional inverse of T_k. Moreover $[x^n]\,H = \binom{n-2}{2}$ for $n \geq 4$.*

Proof. Substituting T_d into T_c and multiplying the numerator and the denominator by $1 - dx$,

$$T_c(T_d(x)) = \frac{x/(1-dx)}{1 - c\,x/(1-dx)} = \frac{x}{1 - dx - cx} = T_{c+d}(x).$$

(Each step is an identity of power series, since all denominators have constant term 1.) The other statements follow: $T_1^{\circ k} = T_{1+\cdots+1} = T_k$ for $k \geq 0$, and $T_k \circ T_{-k} = T_0 = x$. For H: the coefficient of x^j in $(1-x)^{-3} = (1 + x + x^2 + \cdots)^3$ is the number of ways to write j as a sum of three nonnegative integers in order, which is $\binom{j+2}{2}$; put $j = n - 4$. □

The map $c \mapsto T_c$ thus turns addition of numbers into composition of series. This is why the iterates of T_1 have the explicit form T_k. (The fractions T_c are special cases of Möbius transformations, whose compositions correspond to products of 2×2 matrices.)

Lemma 4.3. *Let $T = T_1 = x/(1-x)$ and let $T' = 1/(1-x)^2$ be its derivative. Then, as power series with integer coefficients,*

$$H \cdot T'(T) + H(T) = \frac{2x^4}{(1-2x)^3}, \tag{3}$$

$$(1-x)H - x^2\,T - x^3\,T' = -\frac{2x^3}{1-x}. \tag{4}$$

Proof. Since $1-T = \frac{1-2x}{1-x}$, we get $T'(T) = \frac{1}{(1-T)^2} = \frac{(1-x)^2}{(1-2x)^2}$ and $H(T) = \frac{T^4}{(1-T)^3} = \frac{x^4}{(1-x)(1-2x)^3}$. Also $H \cdot T'(T) = \frac{x^4}{(1-x)(1-2x)^2}$. Adding,

$$H \cdot T'(T) + H(T) = \frac{x^4\big((1-2x)+1\big)}{(1-x)(1-2x)^3} = \frac{2x^4}{(1-2x)^3}.$$

For (4), bring the three terms to the common denominator $(1-x)^2$: the numerator is $x^4 - x^3(1-x) - x^3 = 2x^4 - 2x^3 = -2x^3(1-x)$. □

4.3 Proof of Theorem 3.1

We first prove the formula for $k \geq 0$, separately for odd and even r, then extend it to negative k, and finally prove the statement about the order.

Proof for odd r and $k \geq 0$. Work modulo $2r$. We have

$$(1-rx)(1-(r+1)x) = 1 - (2r+1)x + r(r+1)x^2 \equiv 1 - x \pmod{2r},$$

because $2r+1 \equiv 1$, and $r(r+1) = 2r \cdot \frac{r+1}{2}$ is a multiple of $2r$ (here we use that $r+1$ is even). By Lemma 4.2, $T^{\circ r} = T_r$ and $T^{\circ(r+1)} = T_{r+1}$, so

$$x + T^{\circ r} \cdot T^{\circ(r+1)} = x + \frac{x^2}{(1-rx)\big(1-(r+1)x\big)} \equiv x + \frac{x^2}{1-x} = \frac{x}{1-x} = T \pmod{2r}.$$

So T satisfies the reduced equation, and by Lemma 4.1, $F_r \equiv T \pmod{2r}$. Since reduction is compatible with substitution, $F_r^{\circ k} \equiv T^{\circ k} = T_k$ for every $k \geq 0$. □

Proof for even r and $k \geq 0$. Work in $\mathbb{Z}/2r\mathbb{Z}$. There $r^2 = 2r \cdot \frac{r}{2} = 0$, so Lemma 2.2 applies with $\varepsilon = r$. Let $f = T + rH$.

Step 1: $f^{\circ 2} = T_2$. By Lemma 2.2,

$$f(f) = T(T+rH) + r\,H(T+rH) = \big(T(T) + r\,H\,T'(T)\big) + r\,H(T) = T_2 + r\big(H\,T'(T) + H(T)\big).$$

(In the second term, Lemma 2.2 would add $r \cdot r\,H\,H'(T)$, which is 0 because $r^2 = 0$.) By (3) the bracket equals $2x^4/(1-2x)^3$, and $r \cdot 2 = 0$. So $f^{\circ 2} = T_2$.

Step 2: f satisfies the equation. Since r is even, Step 1 and Lemma 4.2 give $f^{\circ r} = (f^{\circ 2})^{\circ r/2} = T_2^{\circ r/2} = T_r = x + rx^2 + r^2x^3 + \cdots = x + rx^2$. By Lemma 2.2 again (with $u = x$, $v = x^2$), $f^{\circ(r+1)} = f(x + rx^2) = f + rx^2 f'$. Multiplying out and dropping every term that contains r^2,

$$f - x - f^{\circ r} \cdot f^{\circ(r+1)} = f - x - (x + rx^2)(f + rx^2 f') = (1-x)f - x - rx^2 f - rx^3 f'.$$

Now $(1-x)f = (1-x)T + r(1-x)H = x + r(1-x)H$, and in the terms already multiplied by r we may replace f by T (the difference is again a multiple of r^2). So the right side equals

$$r\big((1-x)H - x^2T - x^3T'\big) = -\frac{2r\,x^3}{1-x} = 0,$$

by (4) and $2r = 0$. Thus f satisfies the reduced equation, and $F_r \equiv f \pmod{2r}$ by Lemma 4.1.

Step 3: all iterates with $k \geq 0$. If $k = 2j$ is even, $f^{\circ k} = (f^{\circ 2})^{\circ j} = T_2^{\circ j} = T_k$. If $k = 2j+1$ is odd, $f^{\circ k} = f \circ T_{2j} = T(T_{2j}) + r\,H(T_{2j}) = T_k + r\,H(T_{2j})$. Over the integers, $T_{2j} = x + 2jx^2 + \cdots$ agrees with x modulo 2, so $H(T_{2j}) - H$ has even coefficients, and r times it is a multiple of $2r$. Hence $r\,H(T_{2j}) = r\,H$ in $\mathbb{Z}/2r\mathbb{Z}$, and $f^{\circ k} = T_k + rH$. □

Proof for negative k, and of the statement about the order. Write E_k for the series on the right side of the theorem ($T_k + rH$ or T_k). For $k = 2r$ we have shown $F_r^{\circ 2r} \equiv E_{2r} = T_{2r}$, and $T_{2r} = x + 2r\,x^2 + (2r)^2 x^3 + \cdots \equiv x$. So $F_r^{\circ 2r} \equiv x$, and therefore also its inverse $F_r^{\circ(-2r)} \equiv x$. Now let k be any integer, and write $k = j + 2rq$ with $0 \leq j < 2r$ and q an integer. Then $F_r^{\circ k} = F_r^{\circ j} \circ (F_r^{\circ 2r})^{\circ q} \equiv F_r^{\circ j} \equiv E_j$. Finally $E_j \equiv E_k$: the coefficients j^{n-1} and k^{n-1} of T_j and T_k are congruent modulo $2r$ because $j \equiv k$, and j and k have the same parity because $2r$ is even. This proves the formula for every integer k. The coefficient form follows from $[x^n]\,T_k = k^{n-1}$ and Lemma 4.2.

For the order: the coefficient of x^2 in E_k is k, because H starts with x^4. So if $F_r^{\circ k} \equiv x$, then $k \equiv 0 \pmod{2r}$. Conversely, if $2r$ divides k, then k is even, so $E_k = T_k$, and every coefficient k^{n-1} with $n \geq 2$ is a multiple of k, hence of $2r$; so $F_r^{\circ k} \equiv T_k \equiv x$. □

4.4 Proof of Corollary 3.2

Proof of Corollary 3.2. Theorem 3.1 with $r = 2$ gives $A \equiv T_1 + 2H$ and $A^{\circ 2} \equiv T_2 \pmod 4$.

(a) and (b). The coefficients of T_1 are all 1, so $a(n) \equiv 1 + 2h_n \pmod 4$. This is odd, which proves (a). It is 1 when h_n is even and 3 when h_n is odd. The parity of $h_n = \binom{n-2}{2}$ repeats with period 4, because $h_{n+4} - h_n = \binom{n+2}{2} - \binom{n-2}{2} = 4n - 2$ is even, and $h_2, h_3, h_4, h_5 = 0, 0, 1, 3$. So h_n is odd exactly when $n \equiv 0$ or $1 \pmod 4$, which proves (b).

(c). $T_2 = x + 2x^2 + 4x^3 + 8x^4 + \cdots \equiv x + 2x^2 \pmod 4$.

(d). Work modulo 4 and let $f = T + 2H$ be the reduction of A. By Lemma 2.2 with $\varepsilon = 2$ (note $2^2 = 0$ modulo 4), $f^{\circ 3} = f(f^{\circ 2}) = f(x + 2x^2) = f + 2x^2 f'$. Hence, using $(1-x)T = x$ and dropping multiples of 4,

$$f - x\,f^{\circ 3} = (1-x)f - 2x^3 f' = x + 2\big((1-x)H - x^3 T'\big) = x + 2\,\frac{x^4 - x^3}{(1-x)^2} = x - \frac{2x^3}{1-x}.$$

Since $-2 \equiv 2 \pmod 4$, this is $x + 2x^3/(1-x) = x + 2x^3 + 2x^4 + \cdots$. □

5 Checking the results

A mathematical proof can contain mistakes, so the results were checked in two independent ways.

Numerical check. We computed the coefficients of A up to x^{23} exactly from (1). They reproduce all 21 terms published in the entry. Working modulo 8, we then computed A, $A^{\circ 2}$ and $A^{\circ 3}$ up to x^{159} and confirmed statements (a)–(d) of Corollary 3.2 for every $n < 160$. Finally, we confirmed the formula of Theorem 3.1, modulo $2r$, for every $r \leq 8$, every k with $-4 \leq k \leq 9$, and every $n < 90$. The computation was done with the PARI/GP computer algebra system, using code that was written separately from the formal proofs (file `verification/independent_checks.gp` in the accompanying repository [9]). A computation

can only test finitely many cases, so it is a check on the definitions and on the indexing, not a proof.

Formal proof in Lean. Lean [7] is a *proof assistant*: a program in which mathematical statements and their proofs are written in a precise formal language, and every step is checked mechanically down to the basic rules of logic. The theorems of Section 3 are proved in Lean 4 (version 4.19.0) on top of Mathlib [8], Lean's library of formalized mathematics. The statements are collected in the file `lean/Entries/A396099.lean` of the repository, in the namespace `OEIS.A396099`. The general results for the whole family are proved in the folder `lean/Consecutive` of the repository; the last three rows of the table give the full names of these theorems. In Lean, iterates with negative index are built from the compositional inverse, exactly as in Section 2. The table lists where each result of this paper is proved.

Statement in this paper	Lean theorem(s)
Lemma 4.1 for $r = 2$; the series A is normalized and satisfies (1)	`source_existence_uniqueness`, `source_normalized`, `source_equation`
The order: $A^{\circ k} \equiv x \pmod 4$ if and only if 4 divides k ($k \in \mathbb{Z}$)	`exact_composition_order`
Corollary 3.2(a)–(d) for every normalized integer solution of (1)	`a396099_all_four`
Corollary 3.2(a) and (b)	`a396099_odd`, `a396099_residue`, `a396099_period`
Corollary 3.2(c): $A^{\circ 2} \equiv x + 2x^2$, and its coefficients	`a396099_second_iterate_series`, `a396099_second_iterate`
Corollary 3.2(d): $A - xA^{\circ 3} \equiv x + 2x^3/(1-x)$, and its coefficients	`a396099_difference_series`, `a396099_difference`
Lemma 4.1 for every $r \geq 1$, over every commutative ring	`Consecutive.existence_uniqueness`
Theorem 3.1, the formula for every $r \geq 1$ and every integer k	`Consecutive.ziter_solution_mod`
Theorem 3.1, the statement about the order, for every $r \geq 1$	`Consecutive.ziter_solution_mod_eq_X_iff`

The Lean project was rebuilt from source, and the axioms on which each of these theorems depends were listed with Lean's `#print axioms` command. Every one depends only on the three standard axioms of Lean's logic (`propext`, `Classical.choice`, `Quot.sound`); no step is left unproved. A formal proof certifies exactly the statement that Lean checks. Whether that statement says what the OEIS entry means is a separate question, so the Lean definitions were also compared with the definition in the OEIS entry (Section 1.1).

6 Discussion

Scope. Theorem 3.1 determines every coefficient of every iterate of A modulo 4, so it settles all four conjectures of the entry completely. Theorem 3.1 says nothing about remainders modulo numbers that do not divide $2r$, or about the size of the coefficients.

All coefficients are odd. Since 2 divides $2r$, the theorem also determines every F_r modulo 2: the correction rH has even coefficients when r is even, so in all cases $F_r \equiv x/(1-x) \pmod 2$. Thus every coefficient of every F_r is odd.

Higher moduli. A computer calculation suggests that the remainders modulo $4r$ are periodic too. For $r = 1, 2, 3, 4, 5$ and all $n < 130$, the residues of $[x^n] F_r$ modulo $4r$ repeat with period 3, 4, 6, 8 and 3 respectively (from $n = 1$, 4, 1, 2 and 1 on). We have not tried to prove this, and we do not know a formula for the period in general.

A more general equation. The argument for odd r extends as follows. Consider, for any integers $r, s \geq 1$, the equation $F = x + F^{\circ r} \cdot F^{\circ s}$. The proof of Lemma 4.1 works without change, so it has exactly one normalized solution over every commutative ring. Let m be a common divisor of $r + s - 1$ and rs. Then $(1 - rx)(1 - sx) = 1 - (r + s)x + rs\,x^2 \equiv 1 - x \pmod{m}$, and the calculation in the proof for odd r shows that $x/(1 - x)$ solves the equation modulo m. Hence every iterate of the solution is congruent to $x/(1 - kx)$ modulo m. For $s = r + 1$ with r odd this is the case $m = 2r$ of Theorem 3.1; another example is $r = 1$, where $F = x + F \cdot F^{\circ s}$ gives $F \equiv x/(1 - x) \pmod{s}$. We checked this remark by computer for all $r \leq 7$ and $s \leq 8$ with $m > 1$ (38 pairs), for $-3 \leq k \leq 6$ and all $n < 50$. It is not part of the Lean formalization.

Related work. Normalized power series with composition as the operation form a *group*: composition is associative, x is a neutral element, and every element has an inverse (Lemma 2.1). When the coefficients lie in the field $\mathbb{F}_p$, this group is the *Nottingham group* of $\mathbb{F}_p$, which has been studied extensively in group theory; in particular, its elements of finite order have been investigated in detail [3]. Theorem 3.1 gives, for each r, an explicit element of order exactly $2r$ in the group of normalized series with coefficients in $\mathbb{Z}/2r\mathbb{Z}$; for $r = 1$ this group is the Nottingham group of $\mathbb{F}_2$.

About this paper

How the work was done. This paper is one of 29 papers on conjectures from the OEIS that came out of an AI research effort directed by Dev D. Goyal [9]. The roles were as follows.

- *Dev D. Goyal* started the project, chose which problems to pursue, asked for proofs, formal verification and revisions, and decided what to publish. Dev D. Goyal did not supply proof strategies or intermediate arguments.
- *ChatGPT* (OpenAI), using the GPT-6 model with Pro reasoning, worked autonomously from those prompts. It searched the OEIS for suitable conjectures, found the proofs, wrote the Lean 4 formalization, and wrote a first draft of this paper.
- *Claude Opus 5.5*, used with Claude Code, checked the work independently. It rebuilt the Lean project from source and confirmed that the theorems listed in Section 5 rely only on Lean's standard axioms; compared the Lean definitions with the OEIS definition; recomputed the sequence with separately written PARI/GP code, compared it with the terms published in the OEIS, and tested every claim of the paper on many terms; and read every step of the proofs. It then wrote the present version of the paper, expanding the explanations and, in places, writing out an argument in more detail or by a more elementary route. The theorems of Section 3 and the proof strategies are ChatGPT's. The Discussion section, including the short arguments in it, was written by Claude Opus 5.5; its observations were checked by computer and are not part of the Lean formalization.

Dev D. Goyal has verified the results and is responsible for this paper.

Review status. This paper has not been peer reviewed, and it has not been submitted to a journal or to the OEIS.

Priority. An OEIS entry that labels a statement as a conjecture shows only that its contributor did not know a proof; it does not show that no proof existed. During the research only limited searches for earlier work were made. No claim of priority is made.

Corrections. Please report any error in the issue tracker of the repository [9], where the Lean sources and the numerical checks are also available.

Acknowledgements

The sequence and its conjectures are due to Paul D. Hanna [1], who also contributed the other members of the family: A396797, A396798, A396807 (the four entries correspond to $r = 2, 3, 4, 5$). The case $r = 3$ (entry A396797) was proved earlier by Shiva Priyan Shanmuga Sundaram [2].

References

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